AI Playbook and Toolkit for Public Health Departments

Time-Series Analysis, Anomaly Detection, and Outbreak Signals

ANL 240

Time-series analysis examines data collected over time. In public health, time patterns are central to surveillance, outbreak detection, syndromic monitoring, wastewater analysis, emergency response, seasonal planning, and program performance. AI-supported anomaly detection can help identify unusual increases, decreases, or pattern changes that may require investigation. This module explains how time-series and anomaly detection methods can support public health without replacing epidemiologic interpretation. Automated alerts may identify signals faster, but signals are not diagnoses or confirmed events. Staff need to understand baselines, seasonality, reporting delays, thresholds, alert fatigue, and the difference between statistical anomalies and public health significance. Learners will practice designing an anomaly detection review plan that includes data quality checks, baseline selection, threshold rationale, human review, and escalation procedures.

Level	applied/foundational
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling

Learning Objectives

- Define time-series analysis and anomaly detection in public health terms.
- Explain baselines, seasonality, trends, reporting delays, and thresholds.
- Distinguish statistical signals from confirmed outbreaks or public health events.
- Identify risks from alert fatigue, data artifacts, delayed reporting, and model drift.
- Design guardrails for AI-supported surveillance alerts.
- Produce an anomaly detection review plan for one public health data stream.

Module Overview

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This module explains how time-series and anomaly detection methods can support public health without replacing epidemiologic interpretation. Automated alerts may identify signals faster, but signals are not diagnoses or confirmed events. Staff need to understand baselines, seasonality, reporting delays, thresholds, alert fatigue, and the difference between statistical anomalies and public health significance.

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Why This Topic Matters for Public Health AI

Time-series methods matter because public health surveillance is often about change. A single count may be difficult to interpret without knowing whether it is higher than expected for that time, place, population, reporting source, and condition. AI-supported anomaly detection can help staff review large volumes of data and prioritize signals that deserve attention.

However, public health time-series data are shaped by reporting systems as much as disease patterns. Weekends, holidays, provider onboarding, lab code changes, EHR outages, weather events, testing availability, and public attention can all change observed data. An anomaly may indicate an outbreak, but it may also indicate a data feed problem or behavior change.

Responsible use requires a workflow that treats alerts as starting points for investigation. Staff must know what data are being monitored, how the baseline was selected, what triggers an alert, what review steps follow, and when a signal becomes an incident, outbreak investigation, or communication issue.

Public Health Example

A syndromic surveillance team may use anomaly detection to monitor emergency department visits for respiratory symptoms. The system may flag a county when visits exceed the expected range for that day and season. The alert could prompt review of age groups, facilities, diagnosis text, lab data, school absenteeism, and local context.

The alert is valuable only if it enters a disciplined review process. A sudden increase may reflect a true outbreak, a coding change, a new facility feed, a holiday effect, or a backlog. The analyst should review data quality, compare related indicators, assess geography and populations, and document whether the signal requires action or continued monitoring.

Technical Deep Dive

Baseline selection is one of the most important technical decisions. A baseline may use several prior weeks, several prior years, a regression model, seasonal decomposition, or more complex algorithms. The baseline should match the public health question. A short baseline may adapt quickly but miss seasonal context. A long baseline may capture seasonality but fail when underlying systems change.

Time-series data often require preprocessing. Analysts may need to address missing days, duplicate submissions, delayed reporting, facility onboarding, outliers, and known holidays. Data corrections can appear as sudden spikes or drops. AI tools should distinguish between raw signals and quality-adjusted signals whenever possible.

Thresholds control alert volume and sensitivity. A very sensitive threshold may detect early signals but generate many false alarms. A conservative threshold may reduce workload but delay detection. Thresholds should be chosen with input from epidemiologists, program staff, and emergency response leads because operational capacity and consequences matter.

Anomaly detection methods vary in complexity. Simple methods may compare counts to moving averages or standard deviations. More advanced methods may use regression, Bayesian approaches, control charts, machine learning, or ensemble models. More complex methods are not automatically better. In public health practice, interpretability, stability, transparency, and workflow fit often matter as much as statistical sophistication.

Alert review should be documented. Staff should record what signal occurred, what data were reviewed, what explanation was considered, what action was taken, and whether the alert was useful. Over time, these review notes can improve thresholds, reduce alert fatigue, and support accountability.

Risks, Failure Modes, and Guardrails

Reporting artifacts. Feed failures, delayed batches, or coding changes can trigger false alerts. Guardrails include data quality checks, feed monitoring, and partner change logs.

Alert fatigue. Too many alerts can reduce attention. Guardrails include threshold tuning, tiered alerting, user feedback, and periodic alert review.

Overreaction to signals. A statistical signal may be mistaken for a confirmed outbreak. Guardrails include defined review protocols and communication approval.

Missed emerging events. Overly conservative thresholds may miss early signals. Guardrails include multiple indicators, sensitivity review, and escalation pathways for staff concerns.

Application to AI-Supported Workflows

Time-series and anomaly detection methods can support predictive dashboards, syndromic surveillance, wastewater monitoring, outbreak intelligence, and emergency preparedness. Generative AI may summarize signal review, but it should include uncertainty and source indicators. Agentic workflows may route alerts or create tasks, but those actions should be limited to defined protocols and human review.

The best AI-supported surveillance workflows combine automation with epidemiologic judgment. AI can help detect changes, but humans determine whether the signal is meaningful, what additional evidence is needed, and what public health action is appropriate.

Reflection Questions

What time-series data stream would benefit from automated signal review?

What baseline is appropriate for the condition, data source, and decision?

Which reporting artifacts could create false signals?

What threshold balances early detection and alert fatigue?

How will staff document review, decisions, and lessons learned?

Practical Exercise

Create an anomaly detection review plan for one public health data stream. Include the monitored indicator, data source, update frequency, baseline approach, known reporting patterns, threshold rationale, alert tiers, review steps, related indicators, documentation expectations, and escalation criteria.

The plan should identify how staff will distinguish statistical anomalies from data quality issues and public health events.

Expected Artifact or Evidence

Anomaly detection review plan

Baseline and threshold rationale

Alert review protocol

Signal documentation template

Expected Assignment

- Anomaly detection review plan
- Baseline and threshold rationale
- Alert review protocol
- Signal documentation template

Knowledge Check

- 1. What is an anomaly in a public health time series?
- 2. Why does seasonality matter?
- 3. What is a common risk of too many low-value alerts?
- 4. Which statement is most accurate?
- 5. What is strong completion evidence for this module?

References and Resources

- CDC National Syndromic Surveillance Program: https://www.cdc.gov/nssp/
- CDC Evaluation Framework: https://www.cdc.gov/evaluation/
- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html